

How do Archaeologists Identify Scarcity and its Solutions?



How do Archaeologists Identify Scarcity and its Solutions?

The challenge for archaeologists is identifying food scarcity in the past: The lack of something is more difficult to identify than its presence. Archaeologists can identify what people ate, using techniques including zooarchaeology and paleoethnobotany, but identifying the quantity of food available is much less clear, since only a small fraction of seeds and a larger, but still fractional, proportions of bone survive archaeologically. Archaeologists thus look to evidence of cultural responses to scarcity to identify potential times and places of scarcity in the past.

One piece of evidence of scarcity is finding items in the human diet that represent fallback or famine foods. An extreme case presented by Gremillion is cannibalism, but more common is the identification of plant foods not eaten regularly or preferably, such as the weeds listed by Gremillion that were eaten occasionally in the American Southwest.

Another line of evidence is the traces that food and nutrient scarcity leaves on the human skeleton. As described by Gremillion, there are durable markers on human teeth and bones that record periods of poor nutrition and dietary stress. These modifications to bones are cataloged by bioarchaeologists, who study human skeletons from archaeological contexts, and when the frequency of these markers at the population level is compared over time and space, it can be possible to identify specific periods of nutritional scarcity.

Archaeological evidence of food storage, as discussed in the previous module, is not a direct marker of food scarcity. Storage, however, is a common response to predictable, periodic food scarcity, so it can be used with other evidence to document social responses to food scarcity. Changes in the capacity or frequency of storage structures provide evidence for changes in either the real or perceived risk of crop failure and food shortages.

Finally, evidence of changes in agricultural risk management suggests that farmers perceived changing risks. The nature of the changes provides insight into the risks faced by communities in the past. For example, a shift toward more drought-tolerant crops suggests that rainfall became less common or predictable. A reduction in the role of perennial plants (such as fruit trees) in favor of annual field crops (such as wheat) is a sensible response to the increased risk of violence and raiding in less secure times. A shift from farming to herding gives communities greater mobility, increasing their ability to leave behind violence and drought alike. Since we can see these shifts in agricultural strategies through archaeological remains, it is possible to infer how communities responded to changing patterns of risk in their landscapes over time.